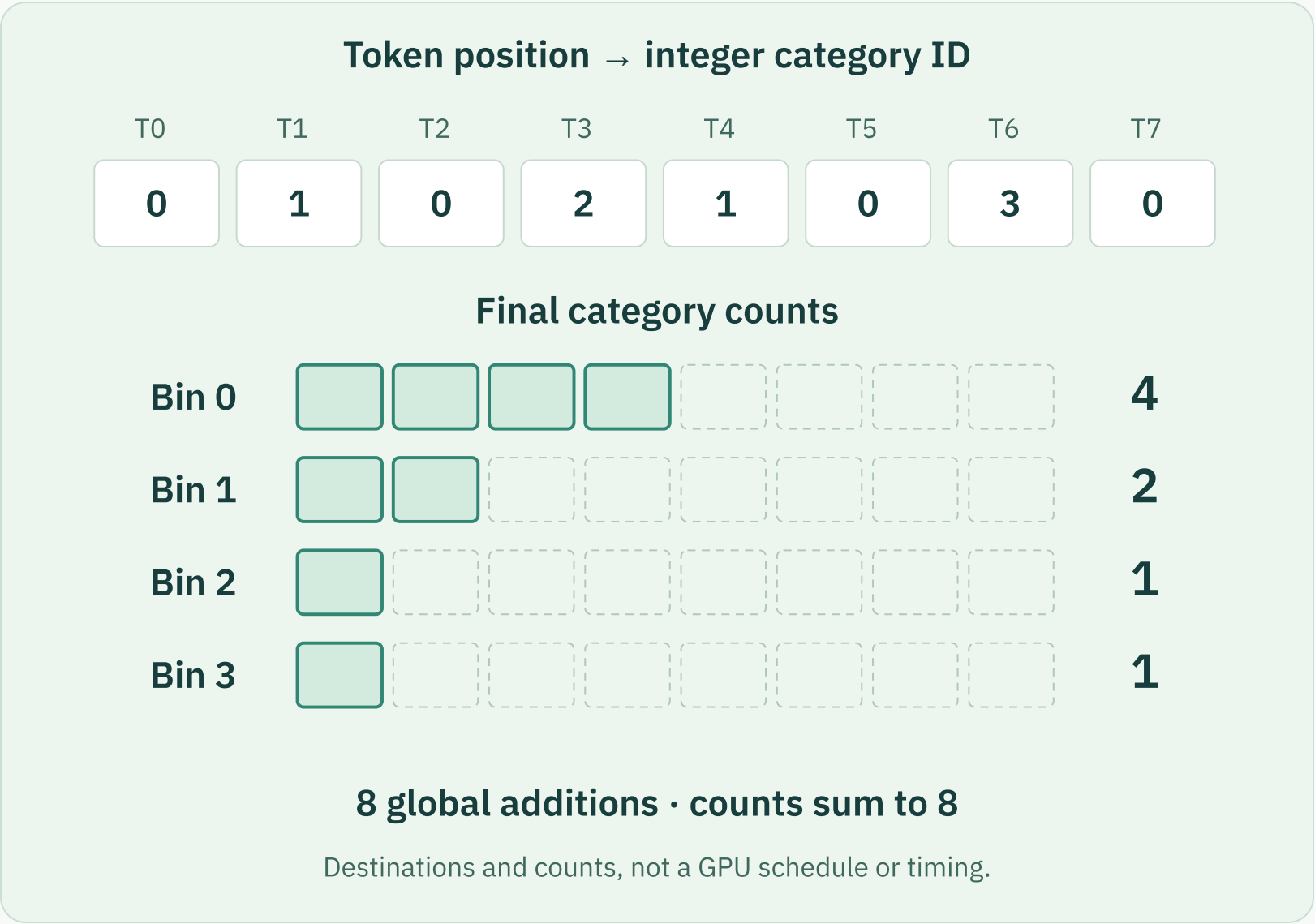
1 / 6

Many tokens can choose one counter.

Eight integer category IDs contribute to four counts.

> Integer category IDs · two four-thread blocks · model operation counts



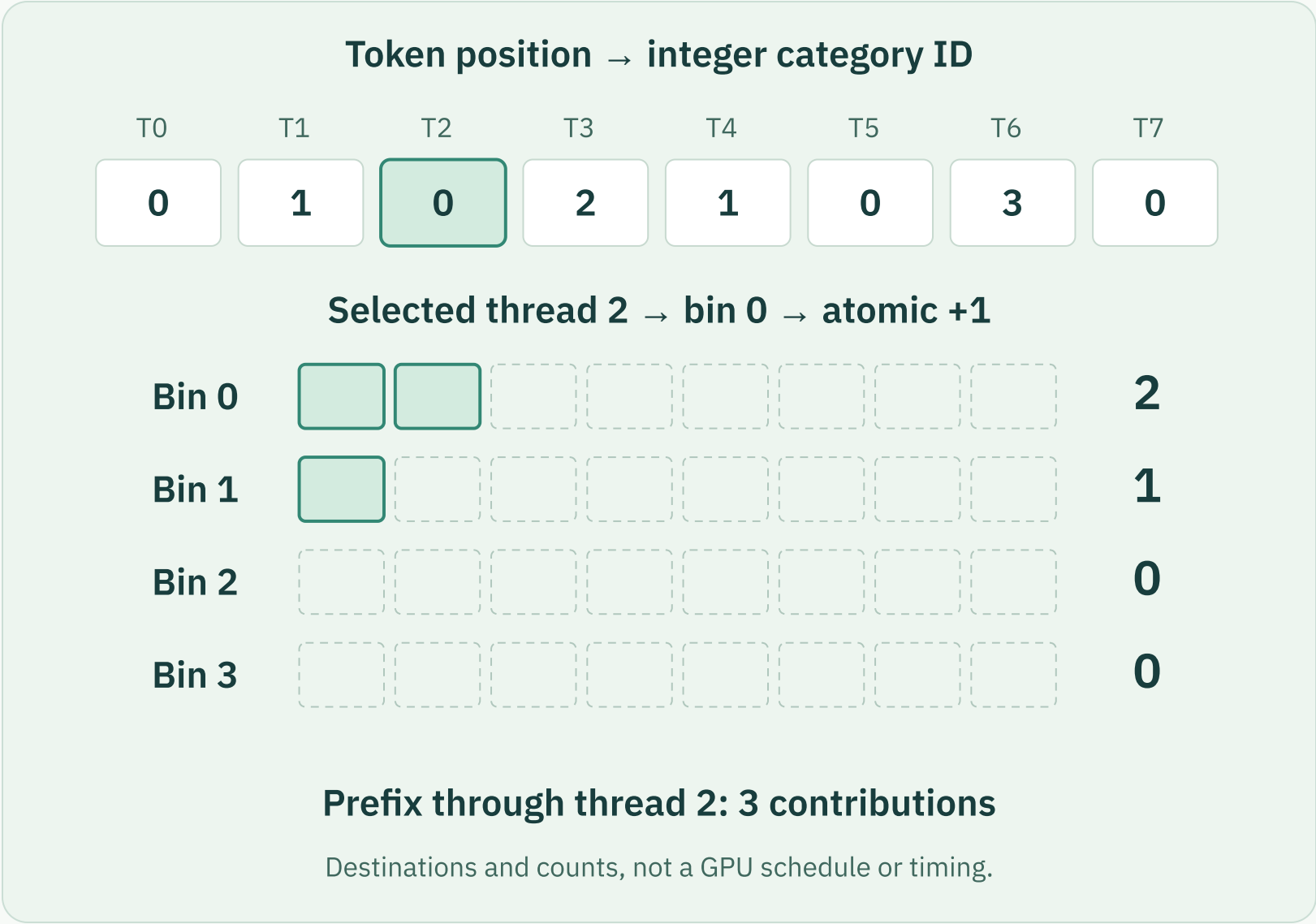
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Protect each shared update.

An atomic addition cannot lose a competing increment.

> Integer category IDs · two four-thread blocks · model operation counts



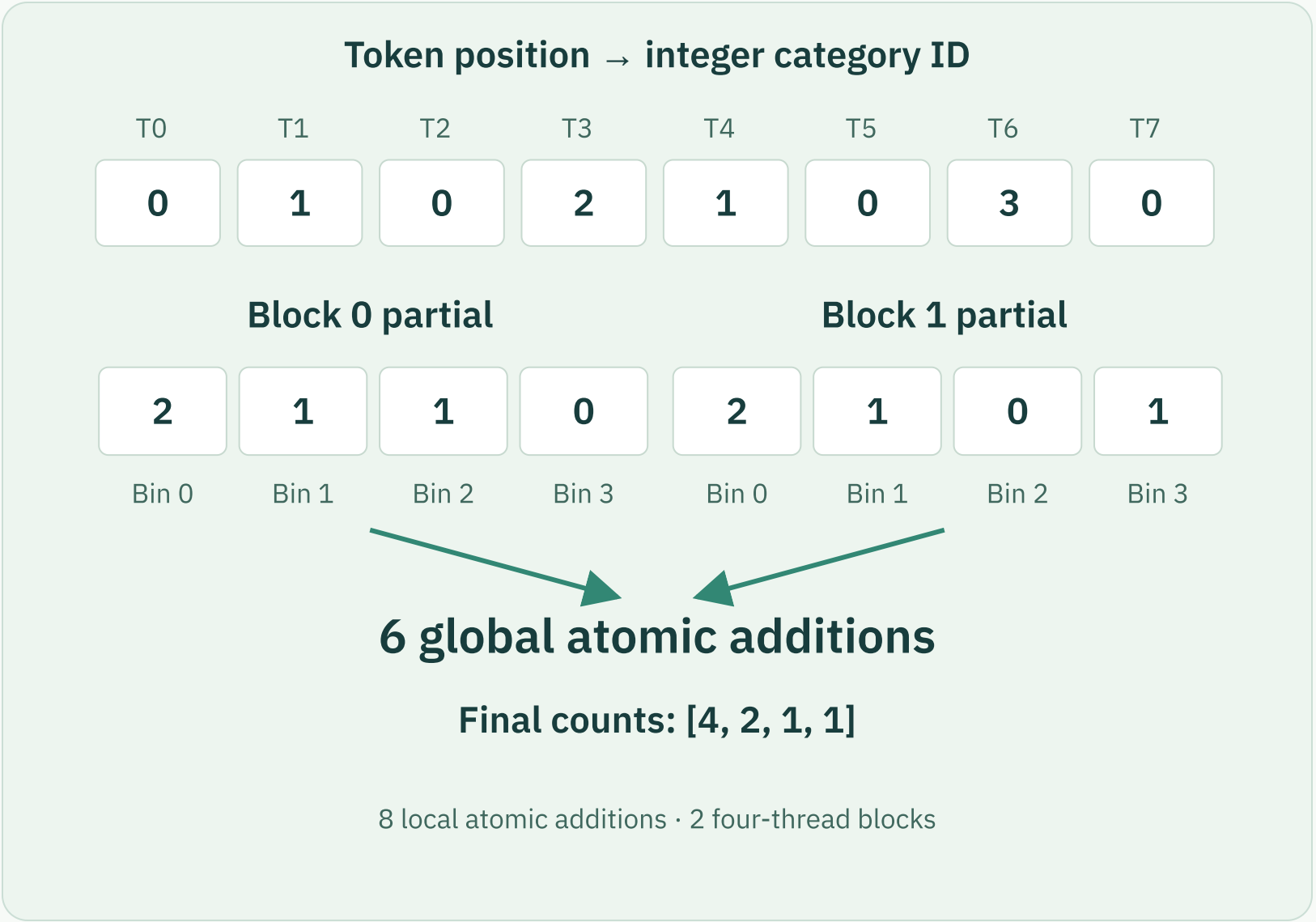
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Count locally, then merge partials.

Two blocks produce separate four-counter histograms.

> Integer category IDs · two four-thread blocks · model operation counts



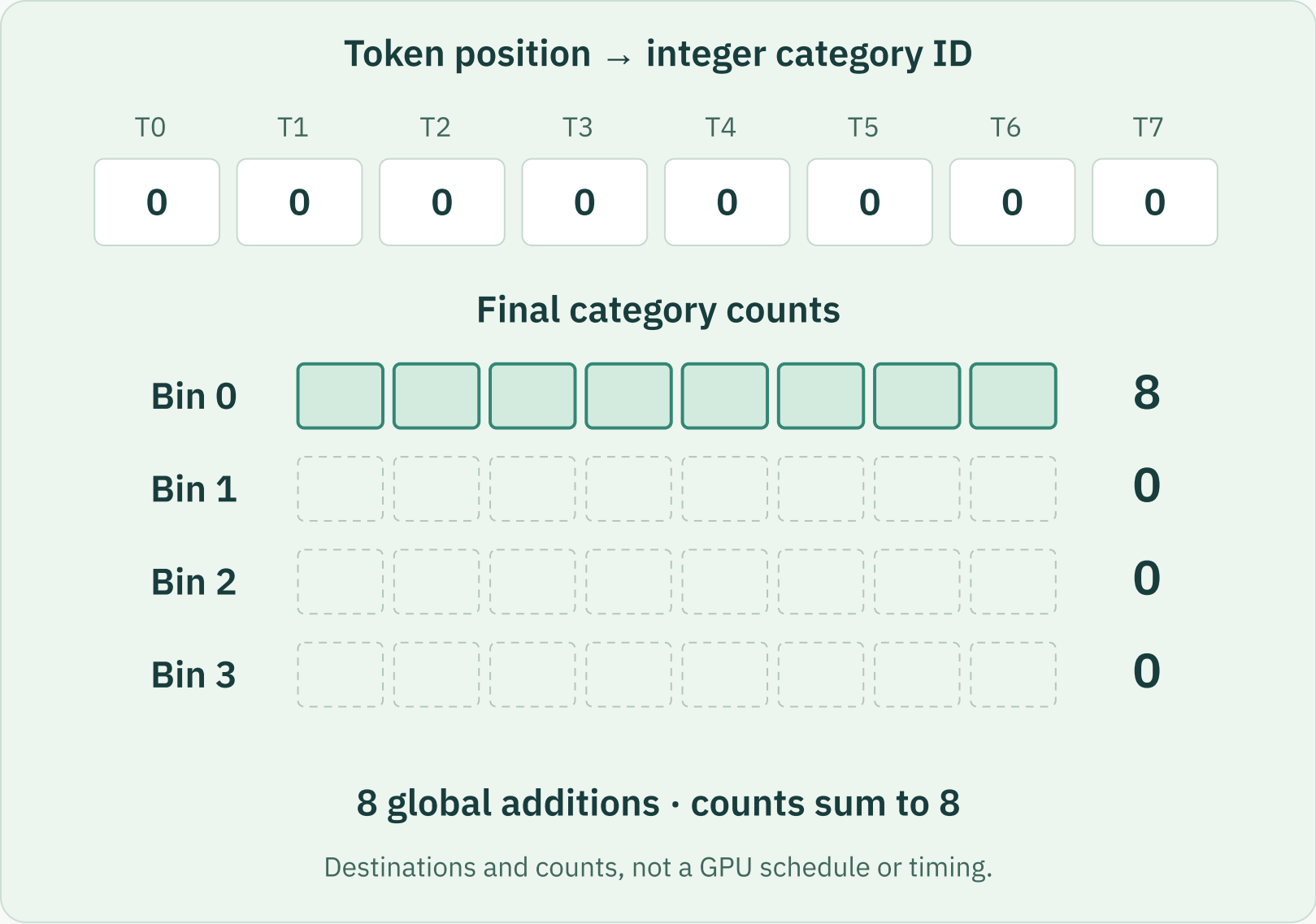
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Concentrated IDs share a destination.

Eight zeros send every update to category zero.

> Integer category IDs · two four-thread blocks · model operation counts



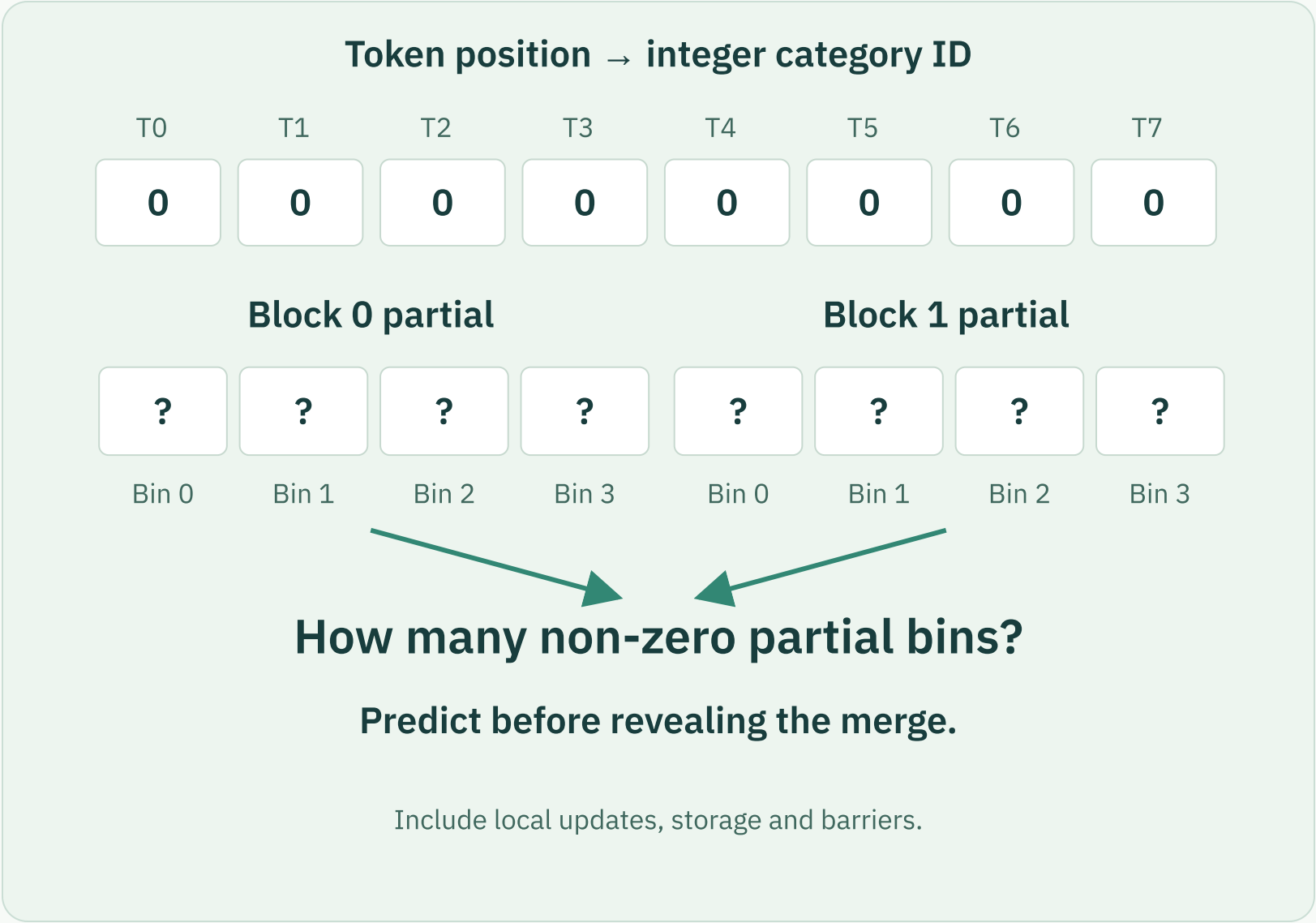
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Pause: change counter ownership.

Keep eight zeros; use one partial histogram per block.

> Integer category IDs · two four-thread blocks · model operation counts



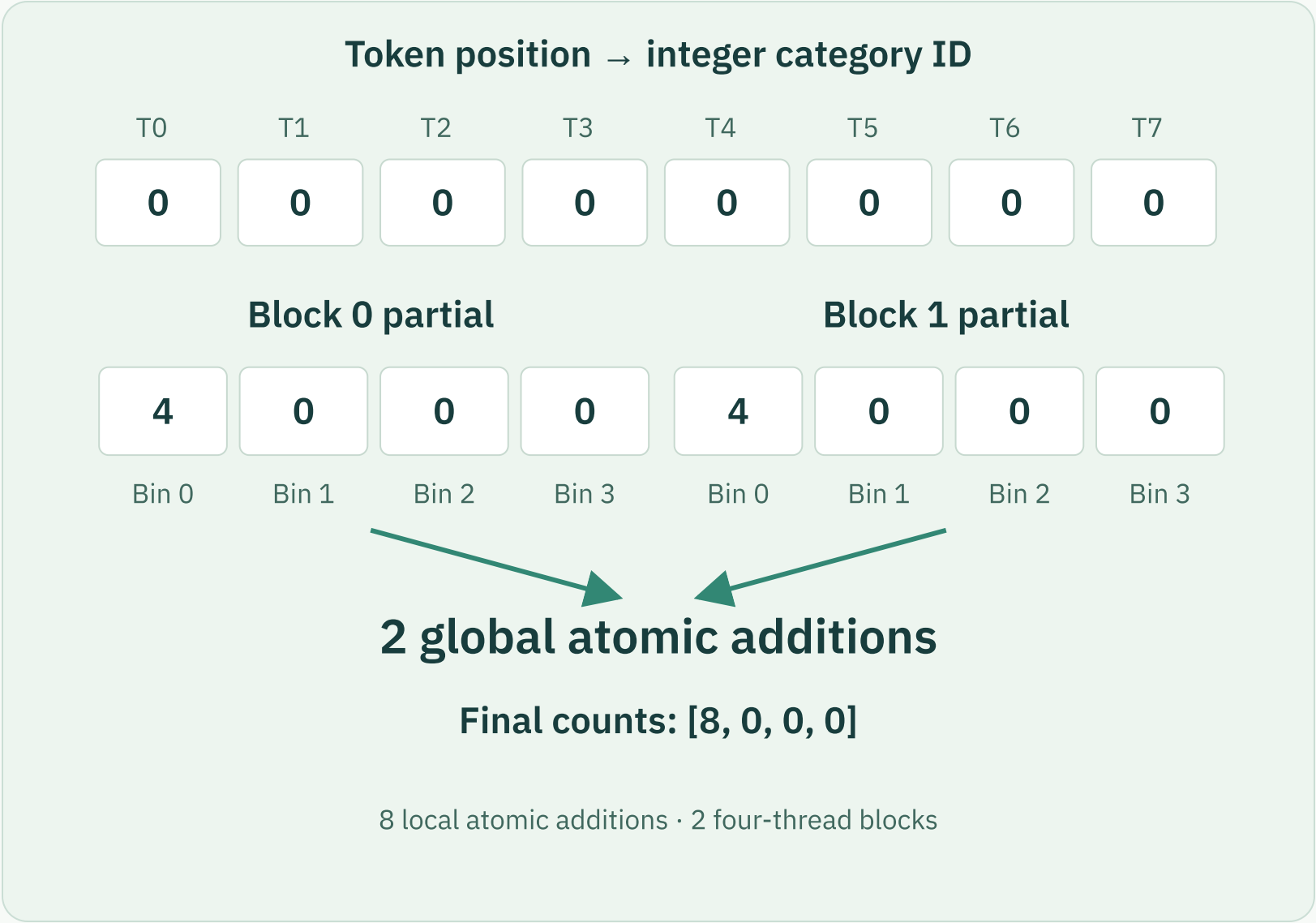
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Two partial contributions reach global memory.

Two global additions still require local work and ordering.

> Integer category IDs · two four-thread blocks · model operation counts



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